

Intuitive Volume Eraser

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Abstract. We present the intuitive volume eraser, an interactive volume rendering tool, to aid direct volume data visualization and Transfer Function (TF) design. Our system adopts sketch-based editing interface which enables interactive exploring and editing operations in an intuitive and natural manner. The data features can be enlightened interactively as user desired. The editing results are saved faithfully with the What You See Is What You Get (WYSIWYG) scheme. We also provide a coupled transfer function editor for users who are used to the traditional TF editing interface. The result of experiments on various data demonstrates the effectiveness of the proposed intuitive volume eraser. The comparison of user experience also shows that our tool outperforms the state-of-the-art approaches in friendliness and efficacy.

Keywords: volume rendering, sketch-based editing, transfer function.

1 Introduction

Direct volume rendering has been widely used to explore three-dimensional volumetric data in different fields. In volume rendering, original volumetric data are often converted to renderable scalar values representing the color and opacity by well-designed mapping functions to highlight the features. Especially, the color mapping is usually performed by transfer function (TF) design. Usually, user would like to see interested features of the original volumetric data. Researchers proposed TF editors to change the color mapping interactively. The desired parts can be visualized or highlighted by hiding or erasing the unwanted parts. However, as surveyed by [1], specifying the TF is a tedious labor work and not intuitive for human perception.

In this paper, we propose an user-friendly volume rendering tool (illustrated by Figure 1), called as intuitive volume eraser, for interactive exploration and visualization of the interested features. With the intuitive volume eraser, users can erase/mask unimportant or uninterested parts directly. Unlike current eraser tools which are indirect and inexact, our erasing tool affords exact cues, like position and data value, and predictive operation. We also provide TF editor for users who are accustomed to the traditional ways. The TF editor provides direct control of color and opacity values and some other

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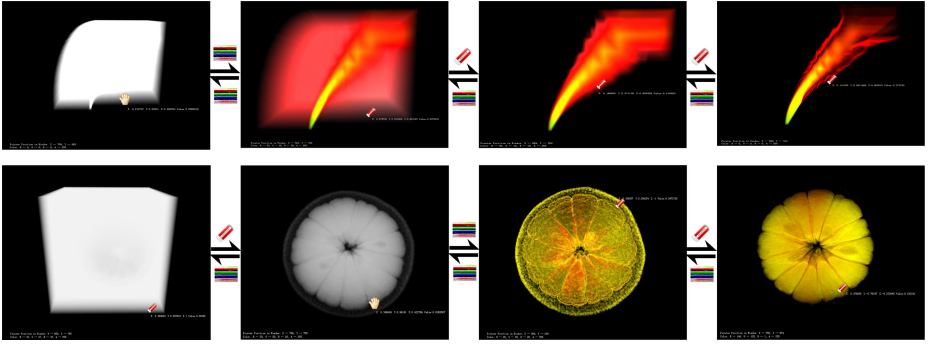


Fig. 1. From left to right: the key features of volumetric data are visualized by the volume eraser in simple way according to user's intuitive interaction. The operations are aided by useful cues with precise position, data value and color information.

useful parameters, such as density and brightness. Consequently, users can easily observe what happens in both volume data space and TF space during erasing operations. This feature is extremely useful for assisting professional TF definition and helping novice to explore and visualize their data.

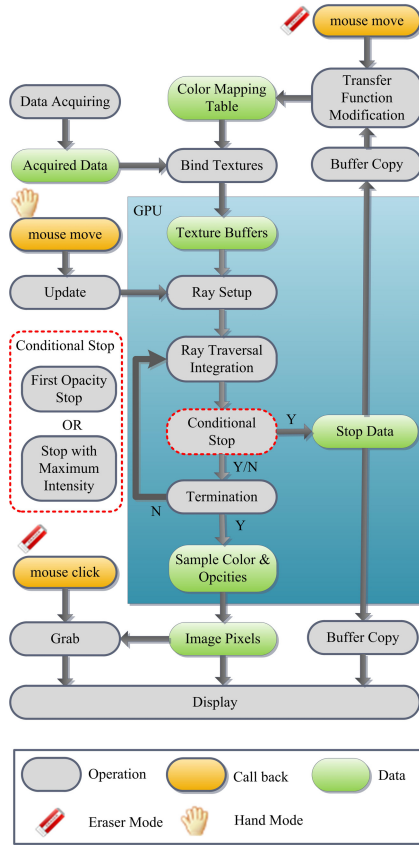
2 Related Work

As the central topic in direct volume rendering, transfer function (TF) has been widely studied in the last decades [1]. According to the utilization of user-interaction, existing TF methods could be categorized into three categories, namely automatic, semiautomatic and interactive.

To realize automatic TF generation, Zhou and Takatsuka [2] utilized topological attributes derived from the contour tree of a volume. The contour tree assisted to look for global features in volumetric data by acting as the visual index of volume segments. Ruiz et al. [3] proposed an automatic approach for TF design without requiring prior knowledge or pre-segmentation information, but need the user provide a target distribution.

Selver et al. [4] proposed a semiautomatic method for initial generation of TFs, allowing users interact with integrated different features. Correa et al. [5] presented semiautomatic methods for generating visibility-driven TFs with the aim to maximize visibility of important features.

Several interactive methods have been proposed to explore volume data more freely. Wu et al. [6] presented a framework which allows users to integrate multiple features and delete features in the direct volume rendering results. Tzeng et al. [7,8] developed an interactive user interface for specifying the classification functions that consist of users painting directly on sample slices of the volume. With VolumeShop [9], Bruckner et al. proposed an interactive system for direct volume illustration. With sketching, interested features are displayed without occluding objects. Guo et al. [10] represented a volume visualization system that accepts direct manipulation through a sketch-based What You See Is What You Get (WYSIWYG) approach. In our work, we present an intuitive tool to erase volumetric data directly.

**Fig. 2.** The volume eraser pipeline with GPU-based ray casting

3 Volume Eraser

To implement our eraser tool, we use Graphics Processing Unit (GPU) based ray casting algorithm as shown in Figure 2. The reasons are two-folds: firstly, ray casting is the most versatile approach for direct volume rendering because of its flexibility in producing high quality images [1]. Secondly, ray casting is apt to parallel execution on GPU to obtain high frame rate, which is vital for interactive operations.

The entire GPU based ray casting process of our system includes the following four steps. Firstly, the source data and color mapping table are acquired by CPU and bound by GPU as textures [1]. Secondly, a ray is setup according to camera parameters and the respective pixel position in the screen. Thirdly, the pixel color and opacity is accumulated by stepping along the ray. Finally, the image pixels are displayed on the screen.

Besides the sampling stops required by the rendering system, we also need to collect stops for eraser. Two kinds of eraser stops are collected correspondingly to two typical erasing strategies. The first one is the first opacity sampling stops which the ray meets.

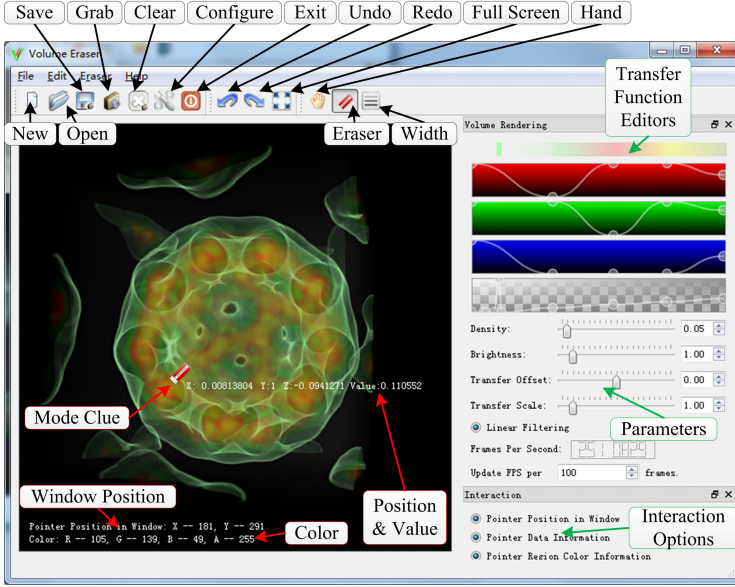


Fig. 3. Screenshot of the intuitive volume eraser tool

The second one is the stop with maximum intensity of all the stops in the ray. As a result, the voxel having the same value with the selected stop will be erased if eraser mode is enabled. If a voxel is erased, its opacity in the TF is set to be zero. It is notable that we only modify the TF without changing the source data.

Fortunately, the required stop information could be collected during the third step of the accumulation process of ray casting. Our rendering and erasing algorithm could be computed in a single pass which is crucial to efficiency. In detail, starting from the eye, the ray comes through certain pixel on the screen and then intersects with the volume. In the sampling and accumulation stage, we choose equidistant steps to benefit front to back traversal. The scalar data value is mapped to optical properties by the color mapping table [1]. The sampling and accumulation proceed along the ray until one of the following termination conditions satisfied: the accumulated alpha α_{dst} reaches a specified threshold (here we choose 0.95); the ray has come through the volume; the number of steps reaches a specified limit (here we use 500). The entire process can be expressed as following equations [1]:

$$C_{dst} = C_{dst} + (1 - \alpha_{src})C_{src} \quad (1)$$

$$\alpha_{dst} = \alpha_{dst} + (1 - \alpha_{src})\alpha_{src} \quad (2)$$

As shown above, we choose opacity-weighted colors [11]. It is worthy to note that the whole ray casting including conditional stops record is implemented by CUDA with GPU.

Figure 3 shows the interface of our tool. A set of system control functions, including setting up new canvas, opening volumetric data file(s), grabbing and saving screenshots,

and interaction modes switching, are provided in the top functional area. The parameter editing area is in the right panel. Users could make use of the transfer function editors and other parameters to refine their visualization results on the fly. The data are visualized and manipulated in the window area. Like most of interface adopting desktop metaphor, we take the mouse as the input device. Manipulate functions are assigned to the left, middle and right button of the mouse. Benefited by the interactive functions, users could manipulate their data precisely and liquidly. Our GPU-based implementation effectively supports our proposed two major editing modes:

- **Hand Mode:** this mode provides a set of fundamental real-time interaction operations, such as rotation, translation and zoom. These operations will result in re-computation of step 2-4 in GPU which could be computed in real-time.
- **Eraser Mode:** in this mode, the volumetric data having the same value with the selected eraser stops will be erased by modifying the transfer function by changing the opacity of this value to 0.0. It is vital to note that the transfer function is updated on the fly while user moves their mouse to erase. The erased rendering result will be displayed in real-time.

Our user interface also provides professional transfer function editors for modifying color and opacity. As depicted in Figure 3, the screen shot of our transfer function editors, from top to bottom, includes integrated transfer function editor for modifying RGBA quadruplets at one fling and four sub-editors for changing red, green, blue and alpha channel respectively. Providing four editors for components of RGBA quadruplets has the following two advantages: Users can recognize and distinguish data values of different colors clearly; More importantly, the data of occluded region can be visible.

4 Results and Discussions

4.1 Results

We implement the intuitive volume eraser using CUDA 4.1 with Qt 4.7.4 on a PC with Intel 2.80 GHz Core i7 CPU, 12 GB RAM and GeForce GTX 580 graphics card.

As shown by Figure 1, the user could gradually operate intuitive volume eraser to produce desirable results. It provides users the functionality to visualize the features directly on volume data space, coupled with the professional transfer function editors. The main property of the eraser is that all these operations are enlightened with the real-time updating cues as discussed in the former sections. In addition, an accompany video also demonstrates the strong functionality of the interactive eraser during volume data exploration and visualization.

Benefited by the GPU-based ray casting implementation, our eraser is performed in real-time. The on-the-fly erasing interaction sequences are shown in Figure 5. The results shown in Figure 4 are achieved in 57 seconds on average for a novice after 5 minutes training. Table 1 illustrates the statistics for these experiments in Figure 4. The statistics indicate that our tool is natural to use and time-efficient. (the FPS of eraser mode is still always larger than 100.) The wide range of dataset also suggests that our tool is broadly applicable.

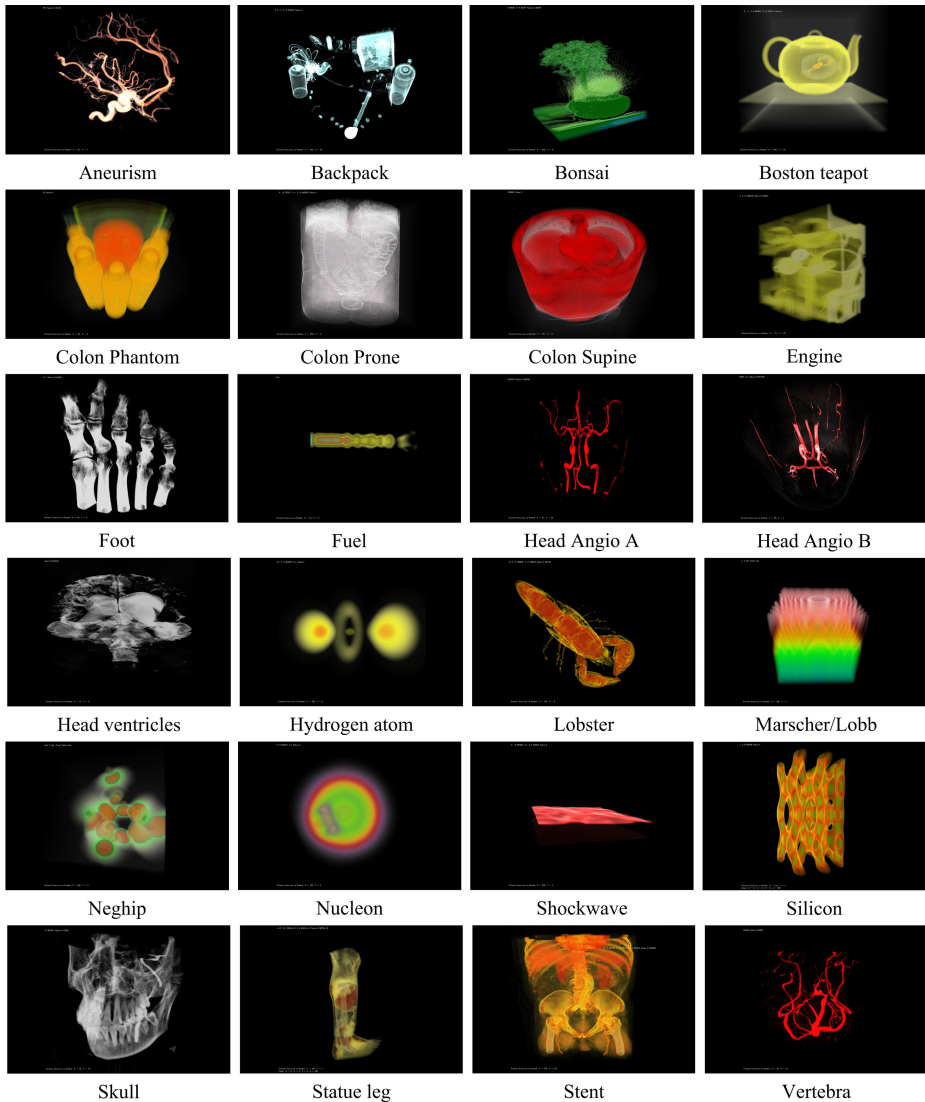


Fig. 4. Evaluation: We tested our tool on the volume datasets one by one. each is finished about 57 seconds for a novice on average.

4.2 Limitations

The visual quality of the basic ray casting algorithm suffers from the issue of interval between sampling stops. Our implementation also effected by this issue. One of our future work would be making use of the recent advances in ray casting to abate the artifacts of our rendering. The fast high-quality volume ray casting method and adaptive sampling could be hopeful candidate.

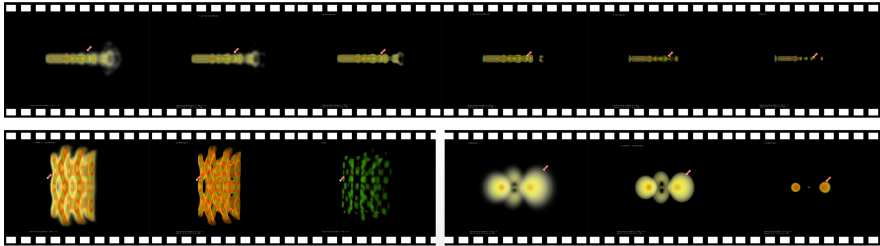


Fig. 5. Snapshots of fluent data filtering. Note that the eraser icon in every picture indicates the position that erasing operation happened. (Top: Fuel; Bottom Left: Silicon; Bottom Right: Hydrogen Atom)

Table 1. Intuitive volume eraser statistics, where frame-per-seconds (FPS) is measured on a PC with Intel 2.80 GHz Core i7 with 12 GB RAM

Dataset	Size	Idle FPS	Interaction FPS	
			Hand	Eraser
Aneurism	256x256x256	195.64	193.50	126.93
Backpack Scan	512x512x373	216.98	207.41	104.45
Bonsai	256x256x256	218.77	217.06	139.61
Boston Teapot	256x256x178	221.21	220.24	153.06
Bucky Ball	032x032x032	232.48	231.03	149.19
Colon Phantom	512x512x442	201.91	191.47	120.48
Colon Prone	512x512x463	220.73	215.19	165.57
Colon Supine	512x512x426	184.40	167.03	123.95
Engine	256x256x128	244.85	232.75	164.69
Foot	256x256x256	202.42	194.62	158.00
Fuel	064x064x064	203.51	195.05	137.71
Head Aneuryism	512x512x512	186.31	170.42	125.64
Head MRI CISS	256x256x124	189.06	182.88	111.38
Head MRT Angiography	256x320x128	205.81	199.94	125.92
Head MRT Angiography	416x512x112	215.88	195.87	114.90
Hydrogen Atom	128x128x128	203.35	215.76	140.68
Leg of Statue	341x341x093	201.46	195.50	117.33
Lobster	301x324x056	200.07	211.76	150.63
Marschner / Lobb	041x041x041	206.48	220.72	155.47
Neghip	064x064x064	197.95	203.41	137.81
Nucleon	041x041x041	201.62	215.56	149.97
Shockwave	064x064x512	214.17	221.18	143.62
Silicon	098x034x034	212.52	226.06	151.16
Skull	256x256x256	222.23	231.32	147.19
Stented Abdominal Aorta	512x512x174	219.53	233.92	146.38

5 Conclusions

In the paper, we present an intuitive WYSIWYG volume eraser that accepts direct sketch on the volume data. Users can explore the data and change the appearance of the direct volume rendered results. The eraser gives real-time feedback to users during the interaction. The experiments on various data demonstrate the performance of our intuitive volume eraser. The comparisons of user experience show that our tool provides more effective and easier-to-use than state-of-art approach for general users.

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